RUSAFI KAMAL

Ottawa, ON | Phone: +1 (613) 857-1663 | Email: rkama064@uottawa.ca | LinkedIn: Rusafi Kamal

SUMMARY OF QUALIFICATIONS

- 3rd Year Mechanical Engineering Co-op student with hands-on experience supporting 80+ engineering projects and mentoring 1,200+ students
- Skilled in CAD modeling, engineering drawings, DFM, and fabrication
- Experience in developing mechanical and mechatronic systems
- Fluent in English and elementary French

EDUCATION

BASc, Mechanical Engineering with Co-op
University of Ottawa (Ottawa, ON) Sep 2024 – 2029

- Represented uOttawa at FYIC 2024 Engineers' conference
- Served as Manager of External Awareness at uOttawa Engineering Students' Society

Highschool Diploma
Bedford Road Collegiate (Saskatoon, SK) Nov 2021 – June 2024

- Saskatoon Public School Proficiency Award (1st place among graduating year)
- 2024 Schulich Leader Nominee

RELEVANT SKILLS

Engineering

- **Mechanical Design:** SolidWorks, Fusion 360, AutoCAD, Onshape, Catia V5, GD&T, Engineering Drawings, DFM, FEA, ASME and ISO Standards
- **Manufacturing:** 3D Printing, CNC Machining, Laser Cutting, Vacuum Forming, Soldering
- **Programming & Controls**: Python, MATLAB, Arduino, MS Visual Studio

Technical

- MS Office (Word, Excel, PowerPoint) | Telemetry/data analysis (AiM logger exports) | Embedded systems (Raspberry Pi, ESP32, Arduino)

Others

- Technical documentation, workshop facilitation & mentoring, leadership, communication

EXPERIENCES

Makerspace Supervisor and Trainer | Bambu Lab™ Campus Ambassador
uOttawa Faculty of Engineering Sep 2025 – Present

- Trained and supported **1,200+** students in CAD design, prototyping, and fabrication workflows
- Hosted **80+** workshops to train students in various manufacturing methods
- Repaired **10+** pieces of manufacturing equipment including 3D printers, laser cutters, and milling machines
- Provided hands-on engineering support for **80+** mechanical and mechatronic projects
- Advised **100+** teams on design for manufacturability (DFM), material selection, and fabrication
- Selected as an on-campus Bambu Lab ambassador out of **500+** applicants, supporting students on the Bambu™ ecosystem and new machines

Summer Camp Supervisor Jul 2025 – Sep 2025
Society of Cumberland (Ottawa, ON)

- Lead a team of 5 co-facilitators overseeing daily activities for **100+** campers, coordinating schedules, activities and safety protocols
- Organized and managed logistics for trips supervising **100+** campers
- Maintained regular parental communication and resolving concerns to ensure camper wellbeing.

PROJECTS

Formula SAE | Aerodynamics Team Member May 2026 – Present

- Analyzed AiM data logger telemetry to diagnose recurring front wing/splitter ground contact
- Implemented **39mm** front ride height adjustment that reduced ground contact incidents by **~80%**
- Conducted structural analysis of rear wing components using FEA to evaluate stress and deformation under loading conditions
- Fabricated rear wing assembly components using workshop manufacturing processes

3-DOF Robot Arm May 2026 – Present

- Designed a robotic arm in CAD, engineering a compact 3-DOF mechanism servo motors and meshing gear trains, supporting a payload of **150g** across a **180°** range of motion per joint
- Manufactured and soldered a PCB for power distribution, integrating battery and charging circuitry
- Self-taught power systems and distribution fundamentals to achieve stable power delivery from battery to mechatronic systems

3D Topology Optimization Manufacturing: Automated CF Layup System Jan 2026 – May 2026

- Designed a passive mechanical feedback loop in SolidWorks to monitor and limit material spooling to a Ø**25mm** threshold via a spring-loaded vertical follower and snap-action trip arm
- Utilized an interlocking dog clutch and dual-bearing shifter fork system in the design to enable positive power transmission and automated drivetrain disengagement
- Optimized system geometry in SolidWorks to achieve a calculated mechanical efficiency **>85%**

Vision-based Assistive System for Object Detection | Honorary Mention Sep 2024 – Dec 2024

- Led a **4-member** team for the development of a computer vision-based obstacle detection system
- Developed and tested an object detection pipeline using Python, OpenCV, YOLO, and Visual Studio, achieving a detection accuracy of **~90%**
- Contributed to system integration between computer software and wearable hardware

Automated Sorting Device Project for CAMECO Corp. Feb 2024 – Jul 2024

- Led a high school design team and built an automated color-based object sorting system
- Integrated a TCS3200 color sensor with servo-driven actuators and embedded controls, achieving ~**95%** accuracy at **15 objects/minute**
- Modeled components in Fusion 360 and generated engineering drawings, ensuring proper tolerance and alignment
- Manufactured a working prototype demonstrating DFM and maintaining proper GD&T